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Patent Search

Invention Title	INTERNET OF THINGS-ENABLED DEVICE AND METHOD FOR ASSISTING USERS IN CULTIVATING AND MAINTAINING PLANTS
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Abstract:

The present disclosure relates to a device (302) for assisting users (106) in cultivating and maintaining plants includes a control unit (116) with processors (202) and a mem (204) storing instructions that, when executed, enable the system to monitor plant health. The device (302) receives sensor data from sensors (304) such as soil moisture, I intensity, temperature, humidity, water level, and pH. The data is processed by comparing it with threshold limits to detect deviations from optimal plant care. Based on deviations, actionable insights are determined using artificial intelligence models, including triggering irrigation, adjusting lighting, or generating alerts. The device (302) ge control signals to manage plant care components (112). The device (302) transmits sensor data and insights to user interface (114) with accessibility features like voice outg vibration alerts to support visually impaired users in managing plant care effectively and independently.

Complete Specification

Description: TECHNICAL FIELD

[0001] The present disclosure relates to a field of Internet of Things (IoT) - enabled indoor plant monitoring and care system. More precisely, the present disclosure relates to an internet of things (IoT) enabled device and method for assisting users in cultivating and maintaining plants.

BACKGROUND

[0002] The following description of the related art is intended to provide background information pertaining to the field of the present disclosure. This section may include certain aspects of the art that may be related to various features of the present disclosure. However, it should be appreciated that this section is used only to enhance the understanding of the reader with respect to the present disclosure, and not as admissions of the prior art.

[0003] Indoor gardening has seen a notable rise in popularity as more individuals pursue sustainable and health-conscious lifestyles. However, while numerous solutions exist for plant monitoring and care, they often overlook the unique needs of visually impaired individuals. Most current systems depend heavily on visual cues such as graphs, charts, and on-screen notifications to convey plant health information. The reliance on visual interfaces excludes users with vision impairments, creating significant barriers for those who wish to participate in indoor gardening independently.

[0004] In addition to the lack of accessibility, existing smart gardening systems frequently suffer from issues of affordability and complexity. Many rely on high-cost components such as Raspberry Pi boards or intricate hardware and software setups that can be daunting for users without a technical background. The affordability and complexity limitations hinder the widespread adoption of such systems, particularly among individuals seeking affordable and user-friendly solutions.

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